MILITARY SPECIFICATION

TEST SET, PRESSURE-TEMPERATURE TTU-205C/E

1. SCOPE

1.1 This specification covers one type of portable pressure-temperature test set, designated TTU-205C/E.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-576	Box, Wood, Cleated, Veneer, Paper Overlaid
PPP-C-843	Cushioning Material, Cellulosic
PPP-C-1120	Cushioning Material, Uncompressed Bound Fiber for Packaging
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-M-7793	Meter, Time Totalizing
MIL-W-16878	Wire, Electrical, Insulated, High Temperature
MIL-T-21200	Test Equipment for use with Electronic and Electrical Equipment, General Specification for
MIL-P-25726	Probe, Total Temperature, Type MA-1
MIL-P-26292	Pitot and Static Pressure Systems, Installation and Inspection of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-470	Maintainability Program Requirements (for Systems and Equipments)

FSC 4920

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MIL-STD-471	Maintainability Demonstration
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-859	Standard Calibration Table for Aeronautical Pressure Measuring Equipment
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MS3102	Connector, Receptacle, Electric, Box Mounting
MS3108	Connector, Plug, Electric, Angle 90 Degree
MS24435	Connector Receptacle, Power Outlet, Electronic Test Equipment

Air Force-Navy Aeronautical

AN6270	Hose Assembly - Detachable Swivel Fitting, Low Pressure

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. Committee on Extension to the Standard Atmosphere

U. S. Standard Atmosphere, 1962

NASA Technical Note D-822 Tables of Airspeed, Altitude, and Mach Number

(Copies of the above publications may be obtained from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 General specification. The requirements of MIL-T-21200, for class 2 equipment, apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

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3.3 Components. The test set shall consist of the following major components and assemblies:

a. One combination case with cover

b. One integral control panel assembly, mounted in the case, incorporating:

(1) A motor driven pressure-vacuum pump

(2) Precision pressure transducer assemblies

(3) Static and differential pressure control systems

(4) Total temperature simulation assembly

(5) Digital altitude and airspeed displays

(6) Operator controls, switches, and indicator lamps

(7) Safety limit devices and warning lamps

(8) Necessary electrical and mechanical parts, wiring, and plumbing

c. One mach number computer, mounted in the case

d. One schematic diagram, mounted in the case

e. One set of operating instructions, mounted in the case

f. One electrical power cable assembly, stored in the case

g. One electrical total temperature simulation cable assembly, stored in the case

h. Three pneumatic hose assemblies, stored in the case.

3.4 Selection of materials, parts, and processes. Selection of materials, parts, and processes shall be in accordance with MIL-T-21200.

3.5 Design and construction. The test set shall be designed for functional operation and use at both organizational and field levels of maintenance. The primary function of the test set shall be to provide regulated total (P_t) and static (P_s) pressure outputs for evaluating aircraft air data systems and equipments. (See note below.) The test set shall be capable of conducting dynamic tests, quantitative calibration tests, and pneumatic-system leak tests. The pressure-regulation system of the test set shall not be affected by ambient barometric pressure changes.

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Note. The two primary pneumatic outputs of the test set are static pressure (P_s) and total pressure (P_t). P_s is atmospheric pressure. P_t is pitot pressure, i.e., the summation of P_s and impact pressure (Q_c), where Q_c is the pressure caused by the movement of the aircraft through the atmosphere. It is recognized that altitude is a direct conversion of P_s, while airspeed is a direct conversion of Q_c. Q_c, for purposes of this specification, shall be P_t-P_s, and will be referred to as differential pressure. All subsequent references to pressures and controls associated with airspeed are stated in terms of differential pressure. References to either airspeed output or total pressure output therefore will be understood to mean P_t output, with P_t referenced to P_s in such a way that P_t-P_s is a direct conversion to airspeed.

3.5.1 General layout. The test set shall be a single, integral unit, self-contained within its case. The case shall serve to house and protect the complete test set and shall have a compartment in the cover for storage of all accessories. All control knobs, switches, and external connections shall be readily accessible and functionally designed. All interior components of the test set, including the pressure-vacuum-pump assembly, shall be so mounted in a single framework attached to the control panel that all interior parts and components can be removed with the control panel, and be completely exposed for maintenance purposes. The test set shall be completely operable when removed from the combination case. The control panel shall have handles to facilitate removal from the case. Detailed mechanical and electrical design of the test set shall be accomplished in accordance with the requirements herein, these requirements being detailed only to the extent considered necessary to obtain the desired mechanical and electrical characteristics, performance, and permanence in accuracy of operation. Design, layout, and assembly of the test set and its component parts shall facilitate quantity production and ease of maintenance.

3.5.2 Electron tubes. Electron tubes shall not be used. Solid state electronic techniques shall be employed to the maximum practical extent.

3.5.3 Input power switch. An on-off switch shall be provided on the control panel for the 115V ac, 400 Hz, single-phase input power to the test set. A simple switch guard shall be provided to protect the power switch from accidental switching. The switch guard shall not lock or completely cover the switch, and shall therefore permit easy access and rapid reaction by the operator if emergency shut-off becomes necessary. A clear indicator lamp shall be provided and shall illuminate to indicate a power-on condition.

3.5.4 Electrical overload protection. The input power switch of 3.5.3 shall be a combination switch and circuit breaker. The circuit breaker shall protect

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all critical circuits and components of the test set during all reasonable conditions, including the incorrect application of electrical power other than that specified in 3.6.1. The circuit breaker shall not be subject to nuisance tripping due to momentary power surges during startup.

3.5.5 Reliability. Reliability of the test set shall be considered of major importance. Reliability effort shall be in effect throughout the design and manufacture of the test set, and shall be achieved via a reliability program established in accordance with MIL-STD-785.

3.5.5.1 MTBF (Mean-time-between-failures). The test set shall have a MTBF of at least 500 hours with a confidence of 90 percent.

3.5.5.2 Longevity. The test set shall have a useful service operating life of 3,000 hours before wearout failures occur or before the equipment consistently fails to meet the MTBF requirements of 3.5.5.1. Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported to the procuring activity.

3.5.6 Maintainability. Maintainability of the test set shall be considered of major importance. Maintainability effort shall be in effect throughout the design and manufacture of the test set, and shall be accomplished in accordance with the design guidelines of MIL-STD-470, paragraph entitled, Establish Maintainability Design Criteria. The guidelines shall be applied wherever applicable, and wherever they do not conflict with other requirements specified herein. Maintenance tasks shall be performed at the lowest practicable maintenance level, using, where practicable, only common handtools and equipment normally available for maintenance of similar equipments.

3.5.6.1 Maintenance downtime. The test set shall have a mean corrective maintenance downtime of 1 hour, and a maximum corrective maintenance downtime of 3 hours, as accomplished by 3 and 5 level repair technicians. Not more than two repair technicians shall be required simultaneously to perform any maintenance task. Periodic overhaul and complete calibration shall be considered as preventive maintenance, and shall not be included as corrective maintenance tasks.

3.5.7 Human factors. The design, layout, and operation of the test set shall be in accordance with basic human factors engineering practices. The human engineering design criteria of MIL-STD-1472 shall be used as a guide, and shall be applied wherever they do not conflict with other requirements specified herein.

3.5.8 Size. The volume of the complete test set, when housed in the combination case with the case cover closed, shall not exceed 3.0 cubic feet. No external dimension shall exceed 25 inches.

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3.5.9 Weight. The weight of the completely assembled test set, including accessories, shall be less than 100 pounds. Weight saving techniques shall be exploited where practicable, and minimum weight shall be a design goal.

3.5.10 Wire. All wire used in the construction of the test set shall be in accordance with MIL-W-16878, type C, D, or E as required, and shall be color coded to facilitate maintenance of the test set.

3.6 Performance. The test set shall be so designed and constructed that no degradation in performance below the requirements of this specification will result when the test set is subjected to the following conditions:

3.6.1 Service conditions, electrical. The electrical power requirements of MIL-T-21200 for class 2 equipment shall apply, with the exceptions and additions specified herein.

3.6.1.1 Frequency. The test set shall not be required to operate from a 60-Hz power source.

3.6.1.2 Current. The test set shall not require a steady state operating electrical current in excess of 4 amps.

3.6.1.3 Electrical safety. The internal input power circuitry shall be so designed that equipment and personnel safety hazards will not exist when the test set is properly connected to the standard aircraft test receptacle of MS24435, or to a standard three-wire laboratory type receptacle. When connecting to a laboratory type receptacle, the appropriate safety provisions shall be in effect with the power source ground on either side of the primary (ie, high or hot) line, and with all test set power cable wires properly connected, including the ground wire. Wiring of the input power circuitry shall be in accordance with the schematic diagram of figure 1 herein.

3.6.2 Service conditions, environmental. The environmental requirements of MIL-T-21200 for class 2 equipment shall apply, with the exceptions and additions specified herein.

3.6.2.1 Electromagnetic interference. The electromagnetic interference requirements of MIL-I-6181 shall apply.

3.6.2.2 Rain. The test set shall be designed to withstand rain in a flight line environment.

3.7 Combination case. The combination case shall be designed in accordance with the requirements of MIL-T-21200. The design objective shall be for minimum weight, consistent with other requirements specified herein. The inside and outside of the case shall be finished as specified for class 2 equipment.

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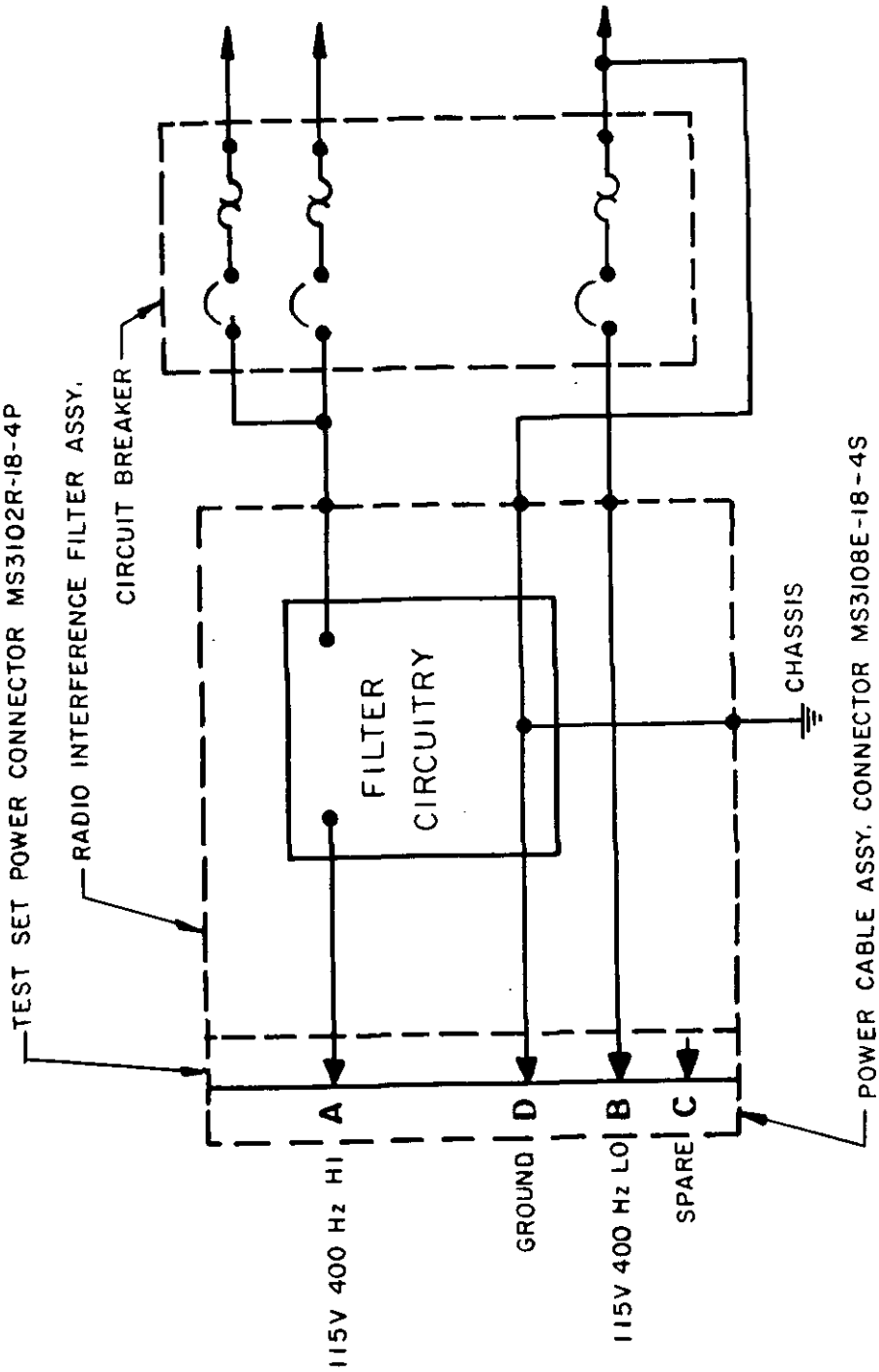


FIGURE 1. Schematic Diagram Input Power Circuitry

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3.7.1 Configuration. The test set shall be integrally housed in the combination case. The case shall be rectangular in shape with the control panel mounted horizontally to be viewed from above. The test set also shall be capable of performing as specified herein when used in a vertical attitude to be viewed from the front. The test set shall be self-supporting in both the horizontal and vertical positions.

3.7.2 Handles. Two metal handles as specified in MIL-T-21200 shall be provided and symmetrically located in the optimum position with respect to weight distribution of the test set. The handles shall be so located that the test set can be carried, without tipping, with the case cover and accessories in place, and with the case cover and accessories removed.

3.7.3 Cover. The case shall be provided with a detachable cover as specified in MIL-T-21200.

3.7.4 Storage compartment. Adequate storage shall be provided in the case cover to store the power cable assembly, temperature simulation cable assembly, and pneumatic hose assemblies. Provisions shall be incorporated in the storage compartment to positively secure these items.

3.7.5 Latches. The case shall be provided with corrosion-resistant steel latches in accordance with MIL-T-21200. The latches shall be a twist-lock type and shall enable quick and easy latching and unlatching without any possibility of injury to the operator.

3.7.6 Control panel. The test set shall incorporate an aluminum primary control panel upon which shall be mounted the pressure controls, function (altitude, airspeed) display indicators, electrical connectors, switches, temperature simulator, warning lamps, output pressure connections, and elapsed time indicator.

3.7.6.1 Elapsed time indicator. A 10,000-hour elapsed time indicator conforming to MIL-M-7793 shall be provided on the control panel, and located in a position readily visible to the operator.

3.7.6.2 Connections. Electrical and pneumatic connections shall be made on the control panel. The connections shall be so situated that the cables and hoses, when connected to the test set, do not interfere with the viewing or operation of the test set when the test set is used in either a horizontal or vertical attitude.

3.7.6.3 Removable subassemblies. The altitude and airspeed digital displays (see 3.9) shall each be integrally packaged with the necessary pressure sensing and transducing mechanisms, so that each subassembly can be readily

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removed and replaced. The subassemblies shall be replaceable from the front of the primary control panel without requiring test set modification or further system calibration. The replaceable altitude and airspeed subassemblies shall be so designed that they can be individually calibrated both installed in, and removed from, the test set (see 3.11.1). When removed from the test set, the subassemblies shall require only a precision pressure source, necessary electrical power, and small common handtools to enable a technician to accomplish calibration.

3.7.6.4 Panel marking. Functional designation of all controls, indications, and connections shall be legibly and durably marked on the control panel in accordance with MIL-T-21200.

3.7.6.5 Panel finish. The color of the primary control panel shall be yellow, and the finish shall be lusterless in accordance with MIL-T-21200. The color of any removable subpanels shall be black, and the finish shall be lusterless.

3.7.6.6 Panel layout. The layout of the control panel shall be as shown on figure 2 herein. Panel mounted components identified by number on figure 2 are listed in table I.

TABLE I. Front Panel Components

Number	Description of Components
1	Total temperature simulation assembly
2	Total temperature connector and cap
3	Power connector and cap
4	Static channel ready lamp (green)
5	Static pressure connector and cap
6	Low altitude warning lamp (red)
7	Static channel ambient vent
8	High altitude warning lamp (red)
9	Negative airspeed warning lamp (red)
10	Pitot channel ready lamp (green)
11	High airspeed warning lamp (red)
12	Pitot pressure connector and cap
13	Mach limit warning lamp (red)
14	Pitot channel ambient vent
15	Vacuum connector and cap
16	Read external mode lamp (amber, neon)
17	Normal/read external toggle switch
18	Normal (generate) mode lamp (clear, neon)
19	Altitude hold lamp (amber)
20	Altitude rate control dial-knob assembly

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TABLE I. Front Panel Components (Cont'd)

Number	Description of Components
21	Cover over air intake and air exhaust
22	Normal/altitude hold toggle switch
23	Airspeed rate control dial-knob assembly
24	Airspeed leak test lamp (amber)
25	Airspeed leak test on/off toggle switch
26	Altitude leak test lamp (amber)
27	Altitude leak test on/off toggle switch
28	Airspeed fine control knob
29	Mach limit set control
30	Altitude fine control knob
31	Mach limit normal/disabled toggle switch
32	Mach limit disabled lamp (red)
33	Modulation amplitude control knob
34	Airspeed coarse control dial-knob assembly
35	Modulation frequency selector switch
36	Power on/off circuit breaker
37	Power on lamp (clear, neon)
38	Elapsed time indicator
39	Airspeed display
40	Altitude coarse control dial-knob assembly
41	Altitude display

3.8 Pressure-vacuum system. The test set shall incorporate a motor-driven pressure-vacuum pump and controller combination having the inherent capabilities of supplying stable static pressures which enable simulation of altitudes from -1,500 to +80,000 feet, and stable total pressures which enable simulation of airspeeds from 50 to 1,000 knots consistent with the test requirements herein. The design shall provide for stable pressure regulation when the system under test has maximum allowable leakage rates as defined in MIL-P-26292 under paragraph entitled Pitot-Static System Leakage, and when the auxiliary vacuum output is connected as specified in 3.8.8.

3.8.1 Motor. The pump motor shall be designed to operate from a 115V ±10 percent, ac, 400 ±20 Hz, single-phase power source. The motor shall be of minimum weight and size consistent with reliability, size, and weight requirements of this specification.

3.8.2 Pump. The combination pressure-vacuum pump may be either a wet or dry type. The inlet of the pressure pump shall incorporate a filter to preclude the entry of foreign materials. The outlet of the pump shall incorporate a filter to preclude the possibility of contaminants, which might inadvertently

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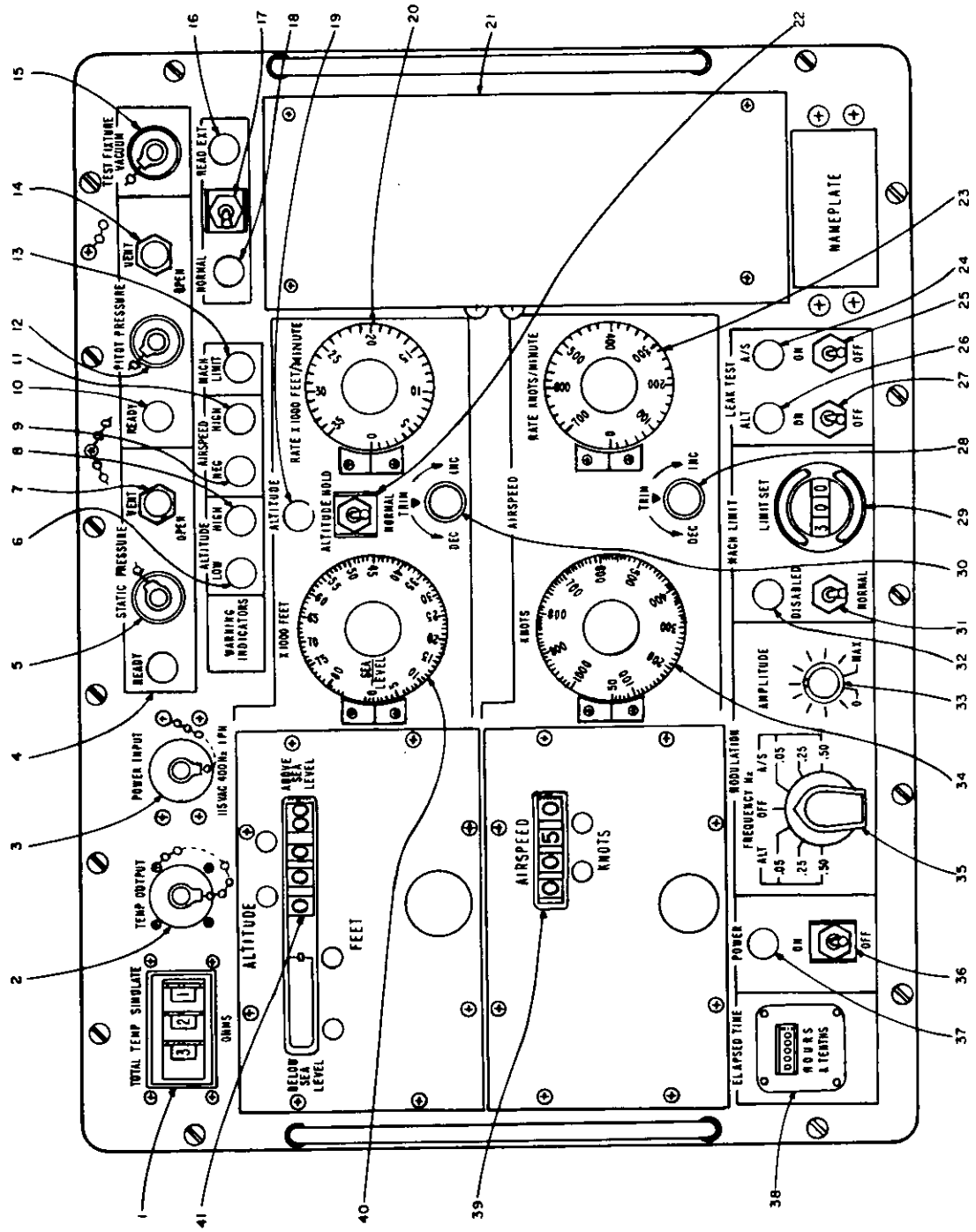


FIGURE 2. Control Panel Layout

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originate within the pump, being passed into the test set pneumatic lines or into the system under test. The pump and drive motor shall be mounted with vibration isolators to prevent excessive vibration from being transmitted to the readout and control instrumentation. The pump shall be of a minimum size and weight consistent with meeting the performance and life requirements specified herein. If a wet-type pump is used, the pump assembly shall be mounted and sealed to preclude loss of lubricant during service use. Provisions also shall be incorporated to facilitate easy draining and filling of the pump assembly without any spillage of lubricant on the internal mechanism of the test set. The pump shall not require a change of lubricant in order to meet the environmental requirements specified herein for both operating and nonoperating conditions.

3.8.2.1 Contamination. The test set shall deliver air free of all contaminants, including oil and water vapor, which would adversely affect the system under test. A moisture removal system shall be provided within the test set to dispose of condensed moisture which may accumulate within the test set.

3.8.2.2 Pump capacity. The pump, when installed in the test set and with the static and total pressure outputs connected to separate external volumes of 250 and 100 cubic inches, respectively, shall have adequate capacity to simultaneously maintain the following static and differential slew rates:

a. Static - At least 35,000 fpm over the entire output range of -1,500 to +80,000 feet

b. Differential - At least 700 knots per minute over the entire output range of 50 to 1,000 knots.

Note. The test set, as specified in 3.8.7.3, must automatically reduce the slew rate, and move slowly into the selected altitude or airspeed to minimize overshoot. Likewise, it is recognized that the test set will require a short period of time to build up to the commanded slew rate, and will not instantly produce the exact required rate. It is therefore understood that although the pump must have adequate capacity to fulfill the requirements of 3.8.2.2, the inherent design of the test set will not permit the maximum slew rates to be maintained throughout the entire test set altitude and airspeed ranges.

3.8.3 Leakage. Provisions shall be included to check the test set, or an aircraft pitot-static system, or both, for excessive leakage. With the static pressure at 50,000 feet and the differential pressure at 700 knots, and the output hoses sealed off, the maximum allowable leak rate of the test set shall not exceed 50 feet and 1 knot per minute, respectively.

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3.8.3.1 Leak test control. The leak test shall be initiated by separate on-off panel mounted controls for the altitude and airspeed outputs. The controls, when placed in the normally OFF position, shall have no effect on the operation of the test set. When placed in the ON position, the controls shall close the leak test valves in the altitude and airspeed output lines, thereby initiating the leak test. An amber indicator lamp shall be provided for each leak test control, and shall illuminate when the respective control is placed in the ON position. Also, when either or both controls are placed in the ON position, the appropriate READY lamp(s) specified in 3.13.6 shall extinguish. The start-cycle-interlock (3.13.6) shall then become effective, and shall prevent pressure surges when the leak test is terminated. Manipulation of the leak test controls under any conditions shall not result in pressure surges or transients in excess of 0.5 inch Hg at the pressure outlets. The test set pump shall continue to operate throughout the leak test.

3.8.4 Static and differential pressure control system. Separate static and differential pressure control systems, each containing a precision pressure transducer, shall be provided to enable an operator to establish static and total pressure outputs to simulate any desired combination of altitude and airspeed within the ranges specified herein. The differential pressure control system shall use the controlled static pressure as its reference pressure in such a way that P_t-P_s is a direct conversion of airspeed. The altitude and airspeed controls shall be manual, and shall each consist of a coarse and a fine adjust control mounted on the control panel. Operation of the controls shall result in a corresponding change in the pressure outputs. The functional displays (see 3.9) shall continuously display the altitude and airspeed outputs being provided by the test set. The controls and displays shall be so designed that the desired values of altitude and airspeed can be quickly and easily obtained.

3.8.4.1 Altitude controls

3.8.4.1.1 Altitude coarse control. The altitude coarse control shall be calibrated directly in feet of altitude, and shall cover the range of -1,500 to +80,000 feet. The control shall be a single revolution type, graduated in increments no greater than 1,000 feet. The altitude coarse control shall not override the zero position of the altitude slew rate control specified in 3.8.7.1.

3.8.4.1.2 Altitude fine adjust. The altitude fine adjust control shall have adequate sensitivity and resolution to enable easy and rapid fine tuning of the desired altitude pressure output with a minimum of manipulation. The control shall have no more than 5 complete revolutions, and shall cover an adjustment range of at least $\pm$500 feet. The altitude fine adjust control shall override

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the zero position of the altitude slew rate control specified in 3.8.7.1, so that the fine adjust function is available at all times while the test set is functioning as a pressure generator in the NORMAL mode (see 3.9.3.1).

3.8.4.2 Airspeed controls

3.8.4.2.1 Airspeed coarse control. The airspeed coarse control shall be calibrated directly in knots of airspeed, and shall cover the range of 50 to 1,000 knots. The control shall be a single revolution type, graduated in increments no greater than 10 knots. The airspeed coarse control shall not override the airspeed slew rate control specified in 3.8.7.2.

3.8.4.2.2 Airspeed fine adjust. The airspeed fine adjust control shall have adequate sensitivity and resolution to enable easy and rapid fine tuning of the desired airspeed pressure output with a minimum of manipulation. The control shall be a single revolution type, and shall cover an adjustment range of at least ±5 knots. The airspeed fine adjust control shall override the zero position of the airspeed slew rate control specified in 3.8.7.2, so that the fine adjust function is available at all times while the test set is functioning as a pressure generator in the NORMAL mode (see 3.9.3.1).

3.8.4.3 Mach limit control. A manually settable control shall be provided on the control panel to limit test set generation of differential and static pressures as a function of mach number. The mach limit control shall be calibrated over a range of 0.80 to 3.00 mach and shall be graduated in increments no greater than 0.05 mach. Accuracy of control shall be ±0.05 mach. The control shall limit the actual mach conditions generated by the test set and shall not be mechanized to close the pressure output isolation valves. The isolation valves shall remain open, and the test set shall continue to generate and provide output pressures as indicated by the altitude and airspeed indicators. The generated pressures, however, shall not exceed the preset mach number regardless of how the altitude and airspeed command controls are manipulated. Operator control of the generated pressures shall be automatically reinstated by changing the commanded altitude or airspeed (or both) to produce a mach number less than the preset limit, or by increasing the preset mach limit to a value greater than that being commanded by the altitude and airspeed controls. Test set operation shall be as follows:

a. A red warning indicator lamp shall be provided to illuminate under the mach limit conditions specified in 3.8.4.3b and c.

b. When an altitude is selected which commands the test set to generate an altitude which would result in a mach number exceeding the preset limit, the test set shall generate the commanded altitude, but shall simultaneously and automatically reduce the generated airspeed as necessary to preclude exceeding the preset mach limit. If the mach limit is reached, the red warning lamp (3.8.4.3a) shall illuminate, and shall remain illuminated as long as the mach limit condition exists.

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c. When an airspeed is selected which commands the test set to generate an airspeed which would result in a mach number exceeding the preset limit, the altitude shall remain as selected, but the test set shall generate only that airspeed which would result in a mach number equivalent to the preset limit. The test set shall not generate airspeeds above this value, even though the commanded airspeed is greater than that being generated. When the excessive airspeed is commanded, the red warning lamp (3.8.4.3a) shall immediately illuminate, and shall remain illuminated as long as the mach limit condition exists, and as long as the excessive airspeed is being commanded.

3.8.4.3.1 Mach limit disable. Provisions shall be included on the control panel to permit an operator to manually disable the mach limit control if mach numbers greater than mach 3.00 are desired. The disable function shall be provided as a separate on-off control located adjacent to the mach limit control. The control positions shall be designated DISABLED and NORMAL, and shall completely isolate the mach limit control function when placed in the DISABLED position. With the mach limit disabled, the test set shall function as specified herein, except that no mach limit shall be imposed on the generation of pressures up to the maximum capability of the test set. A red warning indicator lamp shall be provided to illuminate when the mach limit control is disabled. When the disable control is placed in the NORMAL position, the mach limit control shall function as specified in 3.8.4.3.

3.8.5 Plumbing. The interior plumbing of the test set shall be carefully designed to have a practicable minimum of connections, and to facilitate maintenance. Considerations for choosing the plumbing materials shall include durability, resistance to corrosion and environmental extremes, ability to attenuate vibrations, simplicity of replacement, and minimum weight consistent with other design considerations.

3.8.6 Operating external volume range. The test set shall be capable of proper operation and performance as specified herein with the static pressure output connected to any external volume from 5 to 250 cubic inches, and the total pressure output simultaneously connected to any external volume from 5 to 100 cubic inches.

3.8.7 Slew rate controls. Two separate, manually-operated slew rate controls shall be provided on the control panel to enable an operator to select various combinations of static and differential slew rates within the ranges specified herein.

3.8.7.1 Static slew rate control. The slew rate control for the static output pressure shall permit the selection of any slew rate to simulate altitude rates of change from 0 to 35,000 fpm, either up-scale or down-scale. The control shall be calibrated directly in feet per minute, and shall be graduated in

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increments no greater than 1,000 fpm. The accuracy of the dynamic altitude pressure output at room temperature (25°C) shall be ±150 fpm or ±3 percent of control setting, whichever is greater. The accuracy of the output at temperature extremes (-40° and +55°C) shall be ±300 fpm or ±6 percent of control setting, whichever is greater. The test set dynamic output pressures shall be smooth and free from erratic behavior under any operational conditions specified herein.

3.8.7.2 Differential slew rate control. The slew rate control for the total output pressure shall permit the selection of any differential slew rate to simulate airspeed rates of change from 0 to 700 knots per minute, either up-scale or down-scale. The control shall be calibrated directly in knots per minute, and shall be graduated in increments no greater than 20 knots per minute. The accuracy of the dynamic airspeed pressure output at room temperature (25°C) shall be ±6 knots per minute or ±6 percent of the control setting, whichever is greater. The accuracy of the output at temperature extremes (-40° and +55°C) shall be ±12 knots per minute or ±12 percent of the control setting, whichever is greater. The test set dynamic output pressures shall be smooth and free from erratic behavior under any operational conditions specified herein.

3.8.7.3 Overshoot. The test set, when slewing towards a commanded altitude or airspeed, shall automatically reduce the preset slew rate as the pressures approach the commanded values. This feature shall cause the test set to creep into the commanded altitude or airspeed, thereby minimizing undesirable pressure overshoots. The creep rate, and the point during slew when creep is initiated, shall be determined by optimized trade-offs between overshoot and maintaining the preset slew rate for the longest practicable time during slew.

3.8.7.4 Output/display correlation. The test set altitude and airspeed displays shall accurately follow the dynamic output pressures generated by the test set during slew. Under any slew rate conditions, including maximum slew, the displays shall not differ from the actual output pressures by more than 3,000 feet for altitude and 100 knots for airspeed.

3.8.8 Auxiliary vacuum output. A separate auxiliary pneumatic outlet, designated test fixture vacuum, shall be provided on the control panel to enable a flush-mounted static test fixture to be held, via a vacuum, to the surface of an aircraft equipped with flush-mounted static ports. The outlet shall provide an automatically regulated vacuum output of not less than 4 psi below ambient pressure at any combination of simulated altitude, airspeed, and slew rates specified herein. The vacuum source connector shall be compatible with the AN6270-3 hose assembly specified in 3.15.3. Provisions shall be made in the test set to maintain the vacuum at the required level in the event of electrical power failure, accidental test set shutoff, or shutdown of the test set pump.

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3.8.8.1 Airspeed safety provisions. Automatic safety provisions shall be provided to prevent pressure run-away in the test set airspeed channels if the flush mounted static test fixture is inadvertently removed from the aircraft during test. The test philosophy shall be to prevent damage to both the test set and the aircraft airspeed indicator even though ambient atmospheric pressure is instantaneously introduced into the aircraft static ports, and into the test set static pressure and auxiliary vacuum outlets.

3.9 Functional displays. Two separate functional displays shall be located on the control panel to indicate altitude and airspeed. The displays shall be digital, either electromechanical or electro-optical, with numerals not less than 3/8 inch in height. The mechanization of the displays shall be accomplished by separate static and differential pressure systems (see 3.8.4). When the test set is operating in the NORMAL mode as a pressure generator (see 3.9.3.1), the digital displays shall, at all times while power is applied, display the altitude and airspeed pressures being provided to the system under test. When the internal pressures being generated by the test set are different from the pressures actually being provided to the system under test (ie, when the test set pressure isolation valves are closed), the display indicators shall display the pressures in the output lines, and not the generated pressure (see 3.13.6). Consideration shall be given to providing maximum visibility to the operator when the test set panel is viewed from above (horizontal attitude) or from the front (vertical attitude). The displays shall be illuminated to aid night-time viewing, and shall be clearly readable from the operator's position at a minimum angle of 30° from the normal.

3.9.1 Altitude display

3.9.1.1 Altitude display range. This display shall be calibrated directly in feet of altitude over the range of -1,500 to +80,000 feet.

3.9.1.2 Altitude display accuracy. Accuracy of the altitude display indication shall be achieved without the use of calibration cards. Accuracy of the indication, with respect to the altitude pressure output, shall be as follows:

a. Room temperature (±25°C) - The accuracy shall be ±15 feet or ±0.15 percent of indication, whichever is greater, throughout a range of -1,500 to +60,000 feet, increasing linearly to a maximum of ±0.20 percent at 80,000 feet

b. High temperature (+55°C) - The accuracy shall be ±30 feet or 0.25 percent of indication, whichever is greater, throughout the entire altitude range

c. Low temperature (-40°C) - The accuracy shall be ±40 feet or ±0.30 percent of indication, whichever is greater, throughout the entire altitude range.

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3.9.1.3 Altitude display readability. The readability of the altitude display shall be such that increments of 2 feet may be read directly from the digital display.

3.9.2 Airspeed display

3.9.2.1 Airspeed display range. This display shall be calibrated directly in knots over the range of 50 to 1,000 knots.

3.9.2.2 Airspeed display accuracy. Accuracy of the airspeed display indication shall be achieved without the use of calibration cards. Accuracy of the indication, with respect to the airspeed pressure output, throughout the entire airspeed range, shall be as follows:

a. Room temperature (+25°C) - The accuracy shall be ±1-1/2 knots

b. High temperature (+55°C) - The accuracy shall be ±4 knots

c. Low temperature (-40°C) - The accuracy shall be ±5 knots.

3.9.2.3 Airspeed display readability. The readability of the airspeed display shall be such that increments of 1 knot may be read directly from the digital display.

3.9.3 Indicator mode control. A manually-operated, two-position control shall be provided on the control panel to enable an operator to select either of two operating modes from the altitude and airspeed display indicators. Pressure measurement ranges, accuracies, readabilities, and sensitivities in each mode shall be as specified in 3.9.1 and 3.9.2. The mechanization of the control shall be such that it can be switched to either position, under any operational conditions specified herein, without causing damage to the test set, unit under test, or external pressure generator as appropriate. The switch shall be protected from accidental switching by the same type of switch guard specified for the power switch (3.5.3).

3.9.3.1 Normal mode. When the control is placed in the NORMAL position, the test set indicators shall display the pressures being provided by the test set as specified in 3.9. In this mode, the test set shall function as a pressure generator, and shall perform the pressure generation functions as specified herein. A clear indicator lamp shall be provided to illuminate when the control is placed in the NORMAL position.

3.9.3.2 Read external mode. When the control is placed in the READ EXTERNAL position, the test set pump shall shutdown and the test set shall discontinue the pressure generation function. In this mode, the test set shall function

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as a pressure monitor, and shall be capable of measuring and displaying, by means of the altitude and airspeed indicators, pressures provided to the test set from an external source. The test set shall accept the external pressures through the altitude and airspeed quick-disconnect fittings specified in 3.15.3.1. The pressure relief valve specified in 3.13.1 shall protect the internal test set assemblies from damage by negative differential pressure applied to the test set from an external source. An amber indicator lamp shall be provided to illuminate when the control is placed in the READ EXTERNAL position.

3.10 Pressure modulation capability. The test set shall have the capability to produce, at the static (P_s) and total (P_t) pressure outputs, a modulated sinusoidal pressure wave. Pressure modulation controls shall be provided on the control panel which will enable an operator to select and adjust the frequency and amplitude of the pressure wave, and produce it at either pressure output (not at the same time).

3.10.1 Frequency control. The frequency control shall permit selection of modulation frequencies for either the static or total pressure outputs of 0, 0.05, 0.25, and 0.50 Hz.

3.10.2 Amplitude controls. Separate amplitude controls shall be provided for the altitude and airspeed modulations. The controls shall be continuously variable, and shall be graduated in 10 equal increments with the limits labeled as 0 to MAX. The controls shall permit the selection of any modulation amplitude from zero up to the limit imposed by the test set response characteristics throughout the entire altitude and airspeed ranges. With the control in the MAX position, minimum peak-to-peak amplitudes at 40,000 feet of altitude and 500 knots of airspeed, into 5 cubic-inch volumes, shall be as follows:

Output	Frequency (Hz)	Amplitude
P_s	0.05	1,000 feet
P_s	0.25	200 feet
P_s	0.50	50 feet
P_t	0.05	20 knots
P_t	0.25	1 knot
P_t	0.50	0.2 knot.

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3.10.3 Sine wave distortion. The modulation of the sinusoidal pressure wave shall, in all cases, be symmetrically disposed above and below the unmodulated pressure setting. The test set dynamic output pressures shall be smooth and free from erratic behavior under any operational conditions specified herein.

3.11 Calibration. The altitude pressure outputs shall be calibrated to the U. S. Standard Atmosphere, 1962, as listed in MIL-STD-859. The calibration shall be based on a fixed barometric pressure reference of 29.921 inches Hg. The airspeed pressure outputs shall be calibrated to NASA technical note D-822.

3.11.1 Calibration adjustments. The removable altitude and airspeed subassemblies (see 3.7.6.3) shall each incorporate calibration adjustments, accessible from the front of the subpanel, to enable a technician to accomplish simple routine calibration adjustment without removing the test set mechanism from the combination case. The adjustments shall be protected from tampering, and as a minimum shall enable a technician to shift the entire scale error curve of the particular subassembly, without changing the shape of the curve. Fine calibration adjustments for changing the shape of the curve shall be accessible by removing the subassembly from the test set.

3.11.2 Overall stability of calibration. The overall long term stability of calibration for the entire assembled test set shall be such that recalibration, or replacement of components to maintain calibration, will not be required during a period of at least 90 days or 500 hours of operating time, whichever occurs first. Shifting of the altitude or airspeed scale error curve may be accomplished to meet this requirement (see 3.11.1).

3.11.3 Transducer stability of calibration. The long term stability of calibration of the pressure transducers shall be such that recalibration of the transducers (ie, changing the shape of the scale error curve) shall not be required during a period of at least 180 days or 1,000 hours of operating time, whichever occurs first.

3.12 Operation

3.12.1 Stability of pressure control. After the warmup periods specified in 3.12.3 herein, the test set shall maintain stable pressure control and regulation, while operating into any combination of the external volumes specified in 3.8.6. Stable pressure control and regulation shall be as specified in 3.12.1.1 through 3.12.1.3.

3.12.1.1 Altitude stability

3.12.1.1.1 Normal operation. The normal operating altitude stability (ie, with the altitude hold control of 3.12.1.1.2 in the NORMAL position) shall be

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a maximum altitude variation at room ambient temperature (25°C) of ±6 feet or ±0.03 percent of altitude, whichever is greater, for a period of at least 10 minutes. The normal operating altitude stability at temperature extremes (-40° and +55°C) shall be a maximum altitude variation of ±12 feet or ±0.05 percent of altitude, whichever is greater, for a period of 10 minutes.

3.12.1.1.2 Altitude hold. A manually-operated, two-position control shall be provided on the control panel to enable an operator to select the normal operating altitude stability of 3.12.1.1.1, or to select a more stable altitude hold position. When the control is placed in the NORMAL position, the test set operating functions shall not be affected, and the test set altitude stability shall not exceed the limits specified in 3.12.1.1.1 and 3.12.1.3. When the control is placed in the ALTITUDE HOLD position, the altitude controls specified in 3.8.4.1 shall be overridden, and the test set shall automatically hold the selected altitude pressure. The stability with altitude hold engaged shall be a maximum altitude variation of ±3 feet or ±0.02 percent of altitude, whichever is greater, for a period of at least 30 minutes. The altitude hold stability at temperature extremes shall be ±6 feet or ±0.04 percent of altitude, whichever is greater, for a period of 30 minutes. The airspeed functions of the test set shall not be affected by the altitude hold control. An amber indicator lamp shall be provided to illuminate when the control is placed in the ALTITUDE HOLD position. The switch shall be protected from accidental switching by the same type of switch guard specified for the power switch (3.5.3).

Note. The altitude hold function of 3.12.1.1.2 is intended as an assist in providing the required altitude stability. The altitude hold function is therefore not mandatory, and may be omitted if the stability requirements of 3.12.1.1.2 can be met without providing the altitude hold function. If the stability requirements of 3.12.1.1.2 cannot be met without the altitude hold function, the altitude hold function must be provided.

3.12.1.2 Airspeed stability. The airspeed stability at room temperature shall be a maximum airspeed variation of ±1 knot throughout the entire airspeed range, for a period of at least 30 minutes. The airspeed stability at temperature extremes shall be a maximum airspeed variation of ±2 knots throughout the entire airspeed range, for a period of at least 30 minutes.

3.12.1.3 Oscillation within stability limits. Rapid oscillations, surges, or transients within the stability limits specified in 3.12.1.1 through 3.12.1.2 shall not be allowed. Any instability shall be exhibited as a slow drift. The test set output pressures, when stabilized, shall not drift from dead center to the stability limit (or vice versa) in less than 30 seconds.

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3.12.2 Stability of pressure interaction. The test set, while operating into any combination of the external volumes specified in 3.8.6, shall have minimum interaction between the static and total output pressures. Minimum interaction shall be as follows:

a. At indicator readings of 20,000 feet altitude and 100 knots airspeed, a change in airspeed to 500 knots shall not produce an altitude change in excess of ±20 feet

b. At indicator readings of 2,000 feet altitude and 250 knots airspeed, a change in altitude to 50,000 feet shall not produce an airspeed change in excess of ±2 knots.

3.12.3 Warmup. While maintained at any temperature from -14° to +55°C, the test set shall operate satisfactorily within 10 minutes after power is applied. While maintained at any temperature from -40° to -15°C, the test set shall operate satisfactorily within 20 minutes after power is applied.

3.13 Pressure limit safety devices. Positive means shall be incorporated into the test set to prevent pressure surges, pressure transients, overpressure, and underpressure, both in the test set and the system under test. This shall be accomplished by whatever pressure limiting safety devices are necessary to prevent damage to the test set and the system under test. Under no conditions shall the test set produce pressure surges or transients in excess of 0.5 inch Hg at the pressure outlets. At least the following safety provisions shall be provided:

3.13.1 Negative differential pressure. Safety devices shall be provided to prohibit the static pressure from exceeding the total pressure by more than 0.5 inch Hg. At least two such devices shall be provided, one of which shall be a mechanical pressure relief valve, and the other shall be a pressure actuated electromechanical switch to shutdown the pump as specified in 3.13.5. The pressure switch shall have a brief time delay not exceeding 15 seconds, to prevent nuisance tripping of the pump.

3.13.2 Positive differential pressure. A safety device shall be provided to prohibit the total pressure from exceeding the static pressure by more than 75 inches Hg.

3.13.3 Static pressures. Safety devices shall be provided to prohibit the static pressure from exceeding 32 inches Hg or being less than 0.7 inch Hg.

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3.13.4 Warning lamps. Separate red warning lamps shall be provided, and grouped together in a conspicuous manner on the control panel, to illuminate when the following safety limits have been reached:

a. Mach limit (see 3.8.4.3)

b. Negative differential pressure (low airspeed)

c. Positive differential pressure (high airspeed)

d. High static pressure (low altitude)

e. Low static pressure (high altitude).

3.13.5 Automatic shutoff. Provisions shall be provided in the test set to automatically shutoff the pump if any of the safety limits in 3.13.4b, c, d, and e are reached. The pump shall not shutoff when the mach limit is reached.

3.13.6 Start cycle interlock. An automatic start cycle interlock shall be provided to preclude pressure surges and transients to the system under test during initial starting of the test set. When electrical power is initially applied to the test set, the start cycle interlock shall isolate the system under test from the altitude and airspeed pressures being generated by the test set until the generated test set pressures are within 0.5 inch Hg of the pressures in the system under test. When the generated test set pressures have achieved this pressure equilibrium with the system under test, the test set output lines shall open so that altitude and airspeed pressure outputs can be applied to the system under test. Adjustment of the generated test set pressures to achieve this pressure equilibrium with the system under test shall be accomplished automatically by the test set, and shall not require manipulation of the manual pressure controls. The start cycle interlock shall override the manual controls during the pressure equilization period, and shall automatically reinstate manual control when the test set pressure output lines open. A green indicator lamp designated READY shall be provided for each altitude and airspeed output, and shall illuminate when the respective pressure output line is opened.

3.13.7 Loss of power. The test set shall provide fail-safe operation in the event of inadvertent loss of electrical power to the test set. When electrical power is lost, the fail-safe mechanism shall immediately and automatically isolate and seal the existing test pressures in the system under test (and in whatever portion of the test set output lines is necessary). The start cycle interlock of 3.13.6 shall then become effective, and shall prevent surges when electrical power is reapplied.

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3.13.7.1 Ambient vents. Separate, manually-operated controls shall be provided on the control panel for ambient venting of the altitude and airspeed channels. These vents shall enable the pressures in both the test set and the system under test to be returned to ambient atmospheric pressure if electrical power cannot be restored. The vents shall be so located that the negative differential pressure relief valve specified in 3.13.1 will provide protection for the test set and the system under test while the ambient vents are being operated. The manual controls shall be so designed that the rate of return to ambient can be regulated, but the controls need not be calibrated.

3.14 Total temperature simulation. The test set shall incorporate a precision resistance assembly and connector, wired in accordance with figure 3 herein, which shall be used to simulate the resistance of a total temperature probe of an aircraft at any combination of altitude and airspeed within the ranges and limits specified herein. The resistance range shall be from 30 to 130 ohms, which corresponds to a total temperature range of -99° to +430°C (ie, based on a 50-ohm temperature probe). The total temperature simulation capability shall be fully compatible with aircraft air data computing systems utilizing the three-wire automatic compensation temperature sensing system which enables automatic compensation for the resistance of the cable connecting the test set or temperature sensor to the computer. The precise resistance value set on the test set control dial must appear at test set connector terminals C and D. If the test set is used with a two-wire noncompensated temperature sensing system, the operator must add the resistance of the connecting cable to the value set on the test set control dial to obtain the correct total resistance.

3.14.1 Digital control assembly. A digital control assembly, calibrated directly in ohms, shall be provided on the control panel and shall permit adjustment and control throughout the entire resistance range. The control assembly shall be settable to 0.1 ohm, and the accuracy of any setting at room temperature (25°C) shall be within ±0.1 percent of setting or ±0.05 ohm, whichever is greater. The accuracy at temperature extremes (-40° and +55°C) shall be within ±0.25 percent of setting or 0.15 ohm, whichever is greater. The power rating shall be at least 0.5 watt for any resistance setting. The required resistance range and accuracies shall be available at the test set connector.

3.15 Accessories. The following accessory items shall be provided as part of the test set, and shall withstand the tests specified herein.

3.15.1 Power cable assembly. A primary power cable assembly shall be provided for connecting the test set to an external power source. The cable assembly shall not be permanently attached to the test set. The cable assembly shall be 20 ±0.5 feet in length and shall meet the requirements of MIL-T-21200, except an MS3108E-18-4S connector shall be provided on one end of the cable for connection to the MS3102R-18-4P connector on the test set, and no connector shall be provided on the opposite end of the cable assembly.

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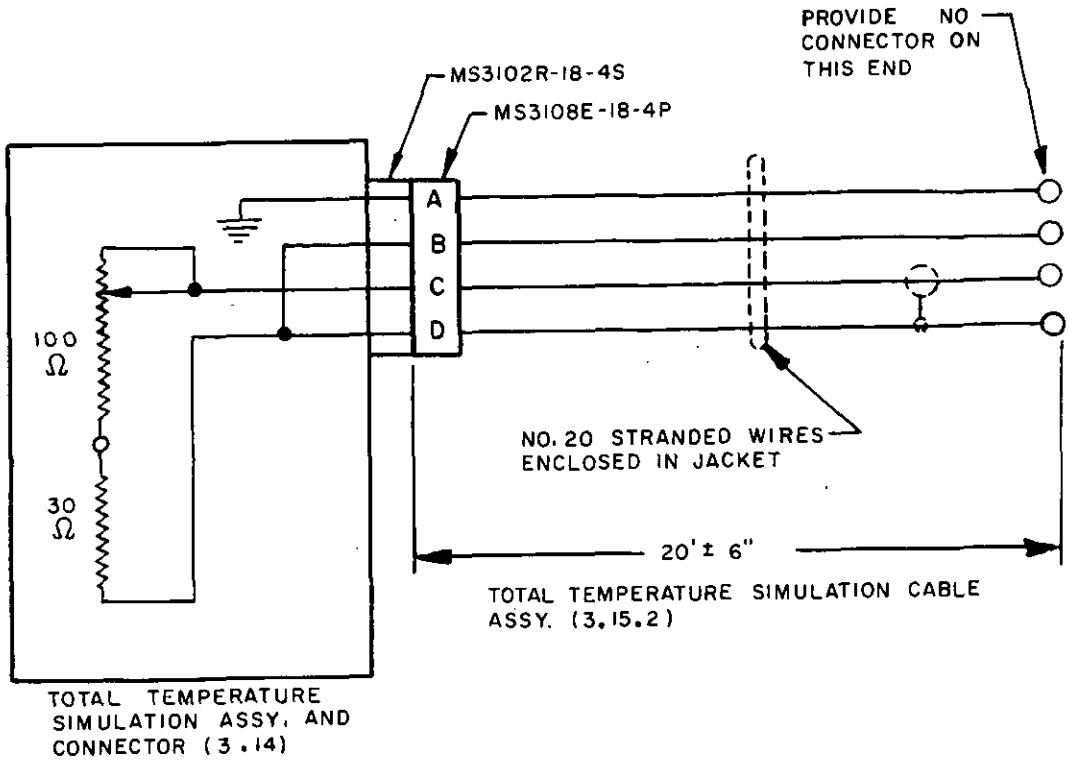


FIGURE 3. Total Temperature Simulation Assembly

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3.15.2 Total temperature simulation cable assembly. A total temperature simulation cable assembly shall be provided for connecting the test set to the equipment under test. The cable assembly shall not be permanently attached to the test set. The cable assembly shall be 20 ±0.5 feet in length, and shall meet the requirements of MIL-T-21200, except that the wiring and connectors shall be in accordance with figure 3 herein.

3.15.3 Pneumatic hose assemblies. Three flexible pneumatic hose assemblies shall be provided for connecting the test set to the equipment under test. The hose assemblies shall not be permanently attached to the test set. The hose assemblies shall be 20 ±0.5 feet in length, and shall meet the requirements of the type AN6270 hose assemblies, except the pneumatic fittings shall be color coded quick-disconnects as specified in 3.15.3.1. Each of the three hose assembly fittings shall be keyed differently to preclude incorrect connection. The sizes of the hoses used in the assemblies shall be as follows:

a. The hose assembly for the auxiliary vacuum output shall conform to AN6270-3

b. The hose assembly for the total pressure output shall conform to AN6270-4

c. The hose assembly for the static pressure output shall conform to AN6270-6.

3.15.3.1 Quick-disconnects. The pneumatic fittings on both ends of the hose assemblies of 3.15.3, and on the corresponding test set pressure outlets, shall be self-sealing quick-disconnects. Upon disconnecting a hose assembly from the test set, the fittings shall instantly and automatically seal both the test set and hose assembly from ambient pressure. It shall be possible to reconnect the hose assemblies without dangerous pressure transients in any of the three pressures. When connected to a mating fitting, the quick-disconnects shall hold a minimum inside to outside differential pressure of 29 inches Hg for 1 minute without leakage. The fittings for the total, static and auxiliary vacuum outlets, and hose assemblies shall be color coded blue, red, and green, respectively, to increase hook-up efficiency.

3.15.4 Mach number computer. A computer shall be provided to enable the operator to compute mach number within 0.02 mach at any point between a range of 0.2 to 3.0 mach based upon readings obtained from the altitude and airspeed indicators. The computer shall be a circular, slide-rule type, no greater than 6 inches in diameter, and equipped with appropriately marked scales. The computer shall be permanently mounted on the inside surface of the cover in a usable position. The computer shall be fabricated from material that will withstand the test conditions specified herein. The design and material of the computer shall be subject to the approval of the procuring activity.

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3.15.5 Operating instructions. A set of operating instructions shall be provided with each test set and secured in the test set compartment. These instructions shall be of a durable nature and shall withstand the tests specified herein. The operating instructions shall concisely outline the procedures to follow in using the test set. Tables showing the conversion of altitude in feet and airspeed in knots to inches Hg for static pressure and differential and total pressures shall also be included in the operating instructions. A table also shall be included showing the conversion of the temperature simulator resistance in ohms to degrees Centigrade and Fahrenheit for the MA-1 total temperature probe as specified in MIL-P-25726.

3.15.6 Diagrams. One wiring diagram and one plumbing diagram, permanent in nature and as large as practicable, shall be mounted on the inside of the test set in a location easily accessible for ready reference during servicing and operation of the test set. The diagrams shall contain pertinent information on the electrical and pneumatic systems of the test set, and shall withstand the tests specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the test set shall be classified as follows:

a. Preproduction testing . . . . . See 4.5

b. Acceptance tests. . . . . See 4.6.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever ambient pressure and temperature for a test are not specified, the test shall be conducted at room ambient pressure and temperature (approximately 25°C).

4.3.2 Test standards. The accuracy of the pressure producing and indicating sections of the test set shall be verified by a comparison with precision pressure standards, the accuracy of which is traceable to the National Bureau of Standards within an accuracy of ±0.003 inch Hg. The accuracy of the total

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temperature simulator shall be verified by using a resistance bridge having an accuracy of at least 0.03 percent of setting. The pressure modulation and slew rate capabilities of the test set shall be verified by using a highly sensitive pressure sensor and recorder. The capability of the test set to limit pressure surges and transients in the outputs shall also be verified in this manner.

4.3.3 Attitude. Unless otherwise specified, the test set shall be tested in a normal operating position (ie, in a horizontal attitude).

4.3.4 Line voltage. Unless otherwise specified, the test set shall be supplied with 115V ac, single-phase, 400-Hz power, with voltage and frequency maintained within ±2 percent.

4.3.5 Allowable warmup time. The allowable warmup time, prior to performing any inspection or measurement, shall not exceed the limits specified in 3.12.3 herein.

4.3.6 Run-in period. Prior to initiation of preproduction and acceptance testing, each test set shall be cycled up-scale and down-scale throughout the entire range of altitude and airspeed for a minimum of two complete cycles. The slew rates, both up-scale and down-scale, shall be 35,000 fpm and 400 knots per minute.

4.4 Test program

4.4.1 Procedures and methods. The procedures and methods for conducting both the preproduction tests and the acceptance tests shall be approved by the procuring activity prior to any formal testing. The procedures shall provide complete and detailed information on the proposed test program including test setups, test methods, and test equipment.

4.5 Preproduction testing (see 6.2)

4.5.1 Test samples. Six test sets representative of production test sets, shall be subjected to the preproduction tests specified in 4.5.3.

4.5.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and furnished to the procuring activity for approval.

4.5.3 Preproduction tests. The preproduction tests shall consist of all tests described under 4.7.

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4.5.3.1 Preproduction test distribution. The distribution of the preproduction tests on the six preproduction test samples shall be as follows:

a. Three of the test samples shall be subjected to all tests described under 4.7 herein, except 4.7.9 (reliability). Each sample need not be subjected to all tests. The tests shall be distributed among the three samples so as to minimize the real time and the cost required to perform the tests. The distribution shall be subject to approval by the procuring activity.

b. The remaining three test samples shall be subjected to the reliability tests described under 4.7.9.

4.6 Acceptance tests. The acceptance tests shall consist of:

a. Individual tests . . . . . See 4.6.1

b. Sampling plans and tests . . . . . See 4.6.2.

4.6.1 Individual tests. Each test set shall be subjected to the following tests as described under 4.7:

a. Examination of product . . . . . See 4.7.1

b. Leakage tests . . . . . See 4.7.2

c. Functional tests . . . . . See 4.7.3

d. Position error . . . . . See 4.7.4

e. Stability. . . . . See 4.7.5.

4.6.2 Sampling plans and tests. Sampling tests shall be performed only when specified by the procuring activity (see 6.2). When sampling tests are specified and not included in the approved test procedures, they shall be performed as specified herein.

4.6.2.1 Sampling plan A. One test set, selected at random from each monthly production lot of 30 or more, or from a 2-month lot if less than 30 are produced in one of the 2 months, shall be subjected to the following tests as described under 4.7:

a. Individual tests . . . . . See 4.6.1

b. High temperature . . . . . See 4.7.7.2

c. Low temperature . . . . . See 4.7.7.3.

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4.6.2.2 Sampling plan B. One test set, selected at random from the first 15 items of each lot of 100 or less produced on the contract or order, shall be subjected to the following tests as described under 4.7:

a. Sampling plan A . . . . . See 4.6.2.1

b. Power variations . . . . . See 4.7.6

c. Low pressure. . . . . See 4.7.7.1

d. Rain. . . . . See 4.7.7.4

e. Humidity . . . . . See 4.7.7.5

f. Fungus . . . . . See 4.7.7.6

g. Salt fog . . . . . See 4.7.7.7

h. Dust . . . . . See 4.7.7.8

i. Explosion . . . . . See 4.7.7.9

j. Vibration . . . . . See 4.7.7.10

k. Shock . . . . . See 4.7.7.11

l. Electromagnetic interference . . . . . See 4.7.8.

4.6.2.3 Sampling plan C. Three test sets, selected at random from the first 15 items of each lot of 100 or less produced on the contract or order, shall be subjected to the reliability test specified in 4.7.9, except longevity testing shall not be required.

4.6.2.4 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.6.2.4.1 Individual tests may continue. For operational and production reasons, individual tests may be continued pending investigation of a sampling test failure, but final acceptance of items on hand or later produced shall not be made until it is determined that items meet all requirements of the specification.

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4.6.3 Defects in items already accepted. Investigation of a test failure could indicate that defects exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.7 Test methods

4.7.1 Examination of product. The test set shall be inspected to determine compliance with this specification with respect to material, workmanship, markings, and general requirements.

4.7.2 Leakage. The pneumatic hose assemblies specified in 3.15.3 shall be connected to the test set, and the hoses shall be sealed airtight at the ends opposite the test set. The test set shall then be energized and the static and differential pressure indicators adjusted by means of the controllers to register 50,000 feet and 700 knots, respectively. When the pressures are stable, the leak test control shall be operated. After a lapsed time of 1 minute, the indications of the static and differential pressure indicators shall not have changed by more than 50 feet and 1 knot, respectively. The leak test indicator lamps shall be observed to function properly.

4.7.2.1 System leakage. The test set shall be connected to accumulators equipped with provisions for introducing known maximum leakage rates as defined for system leakage in the test entitled Pitot-Static System Leakage of MIL-P-26292. The test set shall then be turned on and pressures applied to the accumulators to result in an altitude reading of 50,000 feet and an airspeed reading of 700 knots. The leak test control on the panel shall be operated and the leakage rate noted as registered by the panel indicators. The rates thus established shall be compared with the known leak rates of the external system and shall not differ by more than 100 feet and 2 knots per minute, respectively. Instantaneous release of the leak test controls shall not cause pressure surges exceeding 0.5 inch Hg at the test set pressure outlets. The leak test indicator lamps shall be observed to function properly.

4.7.3 Functional tests. The test set shall be operated to determine that all components are properly assembled and the test set functions properly and meets all performance requirements specified herein. The following minimum tests shall be conducted:

4.7.3.1 Scale error. The static pressure output shall be connected to a 250 cubic inch accumulator, and the total pressure output shall be connected to a 100 cubic inch accumulator, and both outputs shall be connected to the test standards specified in 4.3.2. The static and differential pressure controllers shall be used to obtain each of the altitude and airspeed pressure outputs specified in table II. The values indicated by the altitude and

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TABLE II. Altitude/Airspeed Test Stations

Test Station	Altitude (Feet)	Airspeed (Knots)
1	-1,500	50
2	-1,000	75
3	0	50
4	0	400
* 5	0	100
6	500	150
7	500	500
8	500	200
* 9	1,000	250
10	1,000	600
11	1,000	300
12	5,000	350
13	5,000	700
*14	5,000	400
15	10,000	450
16	10,000	750
17	10,000	500
18	15,000	550
19	15,000	800
*20	15,000	600
21	20,000	150
22	20,000	850
23	20,000	200
24	25,000	250
*25	25,000	900
26	25,000	300
27	30,000	350
28	30,000	950
29	30,000	400
30	35,000	450
*31	35,000	1,000
32	40,000	500
33	40,000	900
34	45,000	550
35	45,000	800
*36	50,000	600
37	55,000	650
38	60,000	150
39	65,000	100
*40	70,000	500
41	75,000	450
42	80,000	400

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airspeed displays for each specified test station shall not differ from the actual altitude and airspeed pressure outputs by more than the limits set forth in 3.9. The accuracies shall be traceable to the National Bureau of Standards via the standard specified in 4.3.2. Readings shall be taken both up-scale and down-scale, but down-scale readings need only be taken at the test points marked with an asterisk. It shall also be shown that the mach limit control functions in accordance with 3.8.4.3, and that the mach limit warning lamp functions properly. The 250- and 100-cubic-inch accumulators shall then be replaced with 5 cubic-inch accumulators and the entire scale error test repeated, except only the even numbered test stations will be required up-scale.

4.7.3.1.1 Read external mode. The indicator mode control switch specified in 3.9.3 shall be placed in the READ EXTERNAL position, and the test set shall function as required in 3.9.3 and 3.9.3.2. The ability of the test set to measure and display externally generated pressures shall be demonstrated at test stations 20 and 31 of table II.

4.7.3.2 Slew rate. The static and total pressure outputs shall be connected to separate 250- and 100-cubic-inch accumulators, respectively, and to a sensor-recorder combination capable of printing a permanent record which can verify that the slew rate performance of the test set is as required in 3.8.2.2 and 3.8.7 through 3.8.7.2 herein. The mach limit control shall be disabled, and the disable mechanism specified in 3.8.4.3.1 shall be observed to function properly. The following tests shall then be performed with the mach control disabled:

a. The rate of change controls for the static and total pressure outputs shall be set to 0 (test station 1 of table III), and the coarse pressure controls shall then be set to 80,000 feet and 1,000 knots, respectively. Under these conditions, the altitude and airspeed displays shall not change. The coarse pressure controls shall then be set back to -1,500 feet and 50 knots, respectively. The fine adjust pressure controls shall be operated, and observed to override the zero setting of the slew rate controls as specified in 3.8.4.1.2 3.8.4.2.2.

b. The rate of change controls shall be set to the values shown for test station 2 of table III. The coarse pressure controls shall then be set to 80,000 feet and 1,000 knots, respectively, and the static and total pressure outputs shall slew up-scale at the selected rates of change, and within the accuracy specified in 3.8.7.1 and 3.8.7.2 herein, until the altitude and airspeed indicators display 80,000 feet and 1,000 knots, respectively. The coarse pressure controls shall then be set back to -1,500 feet and 50 knots, respectively, and the static and total pressure outputs shall slew down-scale at the same selected rates and required accuracies.

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c. The test described in 4.7.3.2b shall be repeated for each test station shown in table III.

d. The 250- and 100-cubic-inch accumulators shall be replaced by 5-cubic-inch accumulators and the slew rate tests of 4.7.3.2a, b, and c shall be repeated, except only at the test stations marked with an asterisk.

TABLE III. Altitude/Airspeed Rate of Change Test Station

Test Station	Altitude Rate (Feet/Min)	Airspeed Rate (Knots/Min)
* 1	0	0
* 2	1,000	700
* 3	3,000	100
4	5,000	200
5	10,000	400
* 6	15,000	600
7	20,000	400
8	25,000	200
* 9	30,000	100
*10	35,000	700

4.7.3.2.1 Reverse slew rates. The same test set-up described in 4.7.3.2 shall be used for the following tests:

a. The static and total pressure outputs shall be adjusted so that the altitude and airspeed indicators display 50,000 feet and 250 knots, respectively. The rate of change controls shall be set to 35,000 fpm and 350 knots per minute, respectively, and the altitude and airspeed controls shall then be simultaneously set to 20,000 feet and 500 knots, respectively. The static and total pressure outputs shall slew at the selected rates of change, and within the accuracy requirements specified in 3.8.7.1 and 3.8.7.2, until the altitude and airspeed indicators display 20,000 feet and 500 knots, respectively.

b. The altitude and airspeed settings shall then be simultaneously reset to 50,000 feet and 250 knots, respectively, and the static and total pressure outputs shall slew to the selected altitude and airspeed at the same rates of change and accuracies specified in 4.7.3.2.1a.

c. The tests described in 4.7.3.2.1a and b shall be repeated for test station 4 of table III.

d. The tests described in 4.7.3.2.1a, b, and c shall be repeated one additional time using 5-cubic-inch accumulators.

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4.7.3.2.2 Pressure surges. While performing the slew rate tests, the test set shall be checked for pressure surges and transients in the pressure outputs. No surge or transient exceeding 0.5 inch Hg shall be produced at the pressure outlets when the various controls are operated. The test set also shall be observed to meet the overshoot and output/display correlation requirements specified in 3.8.7.3 and 3.8.7.4.

4.7.3.3 Safety limit devices. Satisfactory operation of all required pressure and electrical safety limit devices and provisions shall be verified. This shall include all requirements of 3.5.4, 3.8.3.1, 3.8.7.3, 3.8.8, and 3.13 through 3.13.7.1. These tests may be accomplished during scale error and slew rate testing to minimize testing time.

4.7.3.4 Total temperature simulation. The accuracy of the total temperature simulation assembly shall be verified with a suitable resistance bridge as specified in 4.3.2.

4.7.3.5 Pressure modulation. The static and total pressure outputs shall each be connected to 5-cubic-inch accumulators, and to a sensor/recorder combination capable of printing a permanent record to verify the pressure modulation functions specified in 3.10 through 3.10.3. Each output shall be checked individually at all required frequencies specified in 3.10.1, and test station 32 of table II. Minimum wave amplitudes (peak-to-peak) shall be as specified in 3.10.2. Amplitude controls shall be checked for proper operation from 0 to MAX.

4.7.3.6 Auxiliary vacuum output. The performance of the auxiliary vacuum output, as required by 3.8.8 herein, shall be verified with a suitable pressure gage, and shall be monitored throughout the scale error and slew rate tests described in 4.7.3.1 and 4.7.3.2 herein.

4.7.3.7 Pressure interaction. The capability of the test set to meet the pressure interaction stability requirements of 3.12.2 herein shall be demonstrated using external accumulator volumes of 250 and 5 cubic inches, respectively for the static output, and 100 and 5 cubic inches, respectively for the total pressure output.

4.7.4 Position error. The capability of the test set to operate properly in a vertical attitude shall be demonstrated by performing the scale error tests of 4.7.3.1 at test stations 6, 33, and 41 of table II, and the slew rate tests of 4.7.3.2 at test stations 1, 5, and 10 of table III. The tests shall be performed under standard atmospheric conditions, with the static and total pressure outputs connected to 250- and 100-cubic-inch accumulators, respectively.

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4.7.5 Stability. The static pressure output shall be connected to a 250-cubic-inch accumulator, and the total pressure output shall be connected to a 100-cubic-inch accumulator, and both shall be connected to precision pressure measuring and monitoring devices capable of verifying the stability requirements specified in 3.12.1. The following tests shall be performed:

a. The test set shall be adjusted to produce the pressures of test station 5 of table II, and shall be allowed to remain at these pressures for a period of 10 minutes with the altitude hold switch in the NORMAL position. The pressures shall not be adjusted during this period, and shall meet the stability requirements of 3.12.1.1.1, 3.12.1.2, and 3.12.1.3. The test set shall then be placed in the ALTITUDE HOLD mode, and allowed to remain at these pressures for an additional 30 minutes. During this 30-minute period, the test set shall meet the stability requirements of 3.12.1.1.2, 3.12.1.2, and 3.12.1.3.

b. The test specified in 4.7.5a shall be repeated for test stations 31 and 42 of table II.

c. The tests specified in 4.7.5a and b shall be repeated using 5-cubic-inch accumulators.

Note. The tests specified in 4.7.5a, b, and c require continuous operation at each test station for 40 minutes (ie, 10 in NORMAL and 30 in ALTITUDE HOLD). The airspeed stability of 3.12.1.2 may be satisfied during any continuous 30-minute interval during each 40-minute test period.

4.7.6 Power variations. The test set, when supplied with the following power variations, shall operate satisfactorily as specified in 4.7.3 and 4.7.5 herein, except only test stations 1, 5, and 10 of table III shall apply for the slew rate test.

a. 103.5V, 380 Hz

b. 103.5V, 420 Hz

c. 126.5V, 380 Hz

d. 126.5V, 420 Hz.

4.7.7 Environmental tests. The test set, including one complete set of accessories as specified in 3.15 shall be subjected to the following tests conducted in accordance with the specified procedures of MIL-STD-810:

4.7.7.1 Low pressure. The test set shall be subjected to a low pressure test in accordance with method 500, procedure I. The performance of the test set during this test shall be as specified herein, and shall be demonstrated at a

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simulated altitude of 10,000 feet by performing the tests described in 4.7.2, 4.7.3, and 4.7.5 herein, except only test stations 1, 5, and 10 of table III shall apply for the slew rate test. The performance of the test set after this test shall be as specified herein, and shall be demonstrated under standard atmospheric conditions by performing the tests described in 4.7.1 through 4.7.5 herein.

4.7.7.2 High temperature. The test set shall be subjected to a high temperature test in accordance with method 501, procedure I, except the maximum nonoperating temperature shall be 85°C. The performance of the test set during this test shall be as specified herein, and shall be demonstrated at 55°C by performing the tests described in 4.7.2, 4.7.3, and 4.7.5 herein, except only test stations 1, 5, and 10 of table III shall apply for the slew rate test. The performance of the test set after this test shall be as specified herein, and shall be demonstrated under standard atmospheric conditions by performing the tests described in 4.7.1 through 4.7.5 herein.

4.7.7.3 Low temperature. The test set shall be subjected to a low temperature test in accordance with method 502, procedure I. The performance of the test set during this test shall be as specified herein, and shall be demonstrated at -40°C by performing the tests specified in 4.7.2, 4.7.3, and 4.7.5 herein, except only test stations 1, 5, and 10 of table III shall apply for the slew rate test. The performance of the test set after this test shall be as specified herein, and shall be demonstrated under standard atmospheric conditions by performing the tests described in 4.7.1. through 4.7.5 herein.

4.7.7.4 Rain. The test set, in a nonoperating condition and with the case cover removed, shall be subjected to a rain test in accordance with method 506, procedure I. The test set open air vents, both supply and exhaust, shall be suitably covered during this test to prevent excessive deterioration of the internal working test set mechanisms. The performance of the test set after this test shall be as specified herein, and shall be demonstrated under standard atmospheric conditions by performing the tests described in 4.7.1 through 4.7.5 herein.

4.7.7.5 Humidity. The test set, in a nonoperating condition and with the case cover removed, shall be subjected to a humidity test in accordance with method 507, procedure I. The test set open air vents, both supply and exhaust, shall be suitably covered during this test to prevent excessive deterioration of the internal working test set mechanisms. The performance of the test set after this test shall be as specified herein, and shall be demonstrated under standard atmospheric conditions by performing the tests described in 4.7.1 through 4.7.5 herein.

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4.7.7.6 Fungus. If the test set is constructed entirely of fungus inert materials, the contractor may, in lieu of a fungus test, submit a certified statement that no fungus nutrient materials have been used in the construction of the test set. The contractor, if requested by the procuring activity, must demonstrate compliance by submitting a complete list of materials and parts. If fungus nutrient materials are used, fungus testing shall be required and performed in accordance with method 508, procedure I.

4.7.7.7 Salt fog. The test set, in a nonoperating condition and with the case cover removed, shall be subjected to a salt fog test in accordance with method 509, procedure I. The test set open air vents, both supply and exhaust, shall be suitably covered during this test to prevent excessive deterioration of the internal working test set mechanisms. The performance of the test set after this test shall be as specified herein, and shall be demonstrated under standard atmospheric conditions by performing the tests specified in 4.7.1 through 4.7.5 herein.

4.7.7.8 Dust. The test set, in a nonoperating condition and with the case cover removed, shall be subjected to a dust test in accordance with method 510, procedure I. The test set open air vents, both supply and exhaust, shall be suitably covered during this test to prevent excessive deterioration of the internal working test set mechanisms. The performance of the test set after this test shall be as specified herein, and shall be demonstrated under standard atmospheric conditions by performing the tests described in 4.7.1 through 4.7.5 herein.

4.7.7.9 Explosion. The test set shall be subjected to an explosion test in accordance with method 511, procedure I.

4.7.7.10 Vibration. The test set, in a nonoperating condition, shall be subjected to a vibration test in accordance with method 514, procedure X, except the frequency shall be limited to 55 cps. The performance of the test set after this test shall be as specified herein, and shall be demonstrated under standard atmospheric conditions by performing the tests described in 4.7.1 through 4.7.5 herein.

4.7.7.11 Shock. The test set, in a nonoperating condition, shall be subjected to a shock test in accordance with method 516, procedure I, except the shock characteristics shall be as specified in MIL-T-21200. The performance of the test set after this test shall be as specified herein, and shall be demonstrated by performing the tests specified in 4.7.1 through 4.7.5 herein.

4.7.8 Electromagnetic interference. Electromagnetic interference tests shall be performed in accordance with MIL-I-6181.

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4.7.9 Reliability. Three test sets shall be subjected to a reliability test in accordance with MIL-STD-781, test plan XXVI as modified herewith. The environmental test cycle shall be in accordance with test level E of MIL-STD-781, except the low temperature limit shall be -40°C. The performance of the test set throughout the reliability test shall be as specified herein. Performance shall be demonstrated under standard test conditions at 100-hour intervals by performing the tests specified in 4.7.1 through 4.7.5 herein, except the scale error and stability shall be checked only at test stations 5, 20, 31, and 42 of table II, and the slew rate checked only at test station 10 of table III. The final performance check shall consist of all tests specified in 4.7.1 through 4.7.5. The time required to perform the tests, as recorded on the test set elapsed time indicator, shall be counted as part of the reliability test time. During the operating portion of the environmental test cycle, the test set shall continuously slew up-scale and down-scale at an approximately constant rate throughout the entire altitude and airspeed range (ie, -1,500 to 80,000 to -1,500 feet, and 50 to 1,000 to 50 knots). At least two such slewing cycles shall be accomplished for each test set during each environmental test cycle.

4.7.9.1 Reliability accept-reject criteria. Completion of 1,152.5 hours of operation with no failures will permit acceptance. If failures occur, testing shall be continued as outlined in table IV with no more than the indicated number of failures. If calibration of the test set becomes necessary during reliability testing, an additional failure shall be charged against the reliability test. Compliance with table IV will demonstrate a minimum MTBF of 500 hours at a 90-percent confidence level. Failure shall be any condition not in accordance with section 3 herein. Equipment from a production run of which the test group is a part shall not be shipped until an accept decision has been reached on that test group.

TABLE IV. Accept-Reject Criteria

Failures	Total Test Time (Hours)
0	1,152.5
1	1,945
2	2,645
3	3,350
4	4,000
5	4,625
Over 5	Equipment will be rejected subject to redesign as required to demonstrate the specified MTBF.

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4.7.9.2 Longevity. A longevity test shall be conducted in accordance with MIL-STD-781, test plan XXVIII as modified herewith. After the reliability accept criteria have been met, two of the test sets shall continue to be cycled until each test set has accumulated 3,000 operating hours. The test conditions shall be identical to the reliability test, except performance checks need only be made at 200-hour intervals.

4.7.9.2.1 Longevity accept-reject criteria. The longevity accept-reject criteria of MIL-STD-781 shall not apply. Wearout failures shall not occur, and the test sets shall consistently maintain a MTBF of 500 hours throughout the test. The calculation of MTBF shall be as specified in MIL-STD-781 for test plan XXVI. (Five hundred operating hours per equipment). A wearout failure shall be any predictable, time-dependent failure which cannot be corrected by calibration or adjustment.

4.7.9.2.2 Long term stability of calibration. The results of the combined reliability/longevity test shall be used to demonstrate long term stability of calibration as required by 3.11.2 and 3.11.3, using the calibration provisions required by 3.11.1.

4.7.9.3 Replacement of parts. Replacement of major components or assemblies shall not be allowed during the reliability and longevity test. The same major components and assemblies which are in the test set at the beginning of the reliability test shall be used in the test set throughout the reliability and longevity test. Replacement of small detail parts such as resistors, capacitors, valve seals, et cetera, will be allowed if the replacement has been approved by the procuring activity prior to the start of the reliability test, or if the part randomly fails during test. If a failure occurs which necessitates repair of a major component or assembly, the test shall be discontinued (on the failed test set), and the component or assembly repaired. The same component or assembly shall then be placed back into the test set, and testing resumed. If a nonreparable failure occurs to a major component or assembly, the test shall be discontinued and the procuring activity shall be given complete details of the failure. Testing shall not be resumed, unless authorized by the procuring activity.

4.7.9.4 Analysis of test failures. Each failure which occurs during reliability and longevity testing shall be analyzed in accordance with MIL-STD-781.

4.7.10 Maintainability demonstration. Compliance with the maintainability requirements of section 3 herein shall be demonstrated by a maintenance demonstration procedure, maintenance task selection, and maintenance task performance in accordance with MIL-STD-471. Maintenance task selection shall be in accordance with appendix A, and the tasks shall be performed in accordance with appendix B, method 1, test plan pair A_1 plus B_1. Performance of the tasks

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shall be observed by appropriately trained personnel, who shall record maintenance downtimes, tools and equipment required, obstructions or conditions hindering the tasks, and potential hazards to personnel. Depot level maintenance tasks need not be performed, but shall be delineated with a prediction of the downtime and tools required for each task. Signs of excessive wear or other indications of potential test set failure shall be recorded when observed, and the information supplied to the test set design group for study and recommendations. Such recommendations shall be subject to approval by the procuring activity before being implemented.

4.8 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each test set shall be preserved and packaged in accordance with method IIB of MIL-P-116. The case shall serve as the interior container. Desiccant shall be wrapped with kraft paper and positioned inside the case. Handles, corners, and other protrusions shall be padded with cushioning material conforming to PPP-C-843 secured in place with tape conforming to PPP-T-60. The exterior container of the method IIB pack shall be as specified for the applicable level of pack (see 5.2.1 or 5.2.2).

5.1.2 Level C. The test set shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The test set preserved and packaged as specified in 5.1.1 shall be packed in an exterior container conforming to PPP-B-576, style B, class 2. The exterior container shall be lined with a minimum of 1-inch thickness of cushioning material conforming to PPP-C-1120. Containers shall be of uniform size and shape and of minimum tare and cube. Closure strapping shall be in accordance with the appendix of the specification.

5.2.2 Level B. Level B packing shall be the same as level A, except the container shall conform to PPP-B-576, style A, class 1.

5.2.3 Level C. The test set preserved and packaged as specified in 5.1.2 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall conform to either the Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

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5.3 Physical protection. Interior cushioning, blocking, and bracing shall be in accordance with MIL-STD-1186. The edgewise drop tests or cornerwise drop tests shall be in accordance with appendix A to MIL-STD-1186. Tests are not required when level C packing is specified.

5.4 Marking. In addition to any special marking required by the contract or order, interior packages and exterior containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Test Set, Pressure-Temperature TTU-205C/E.

6. NOTES

6.1 Intended use. The TTU-205C/E test set is used primarily at organizational (flightline) and intermediate (field shop) levels of maintenance to provide simulation of static pressure (P_s), total pressure (P_t), and total temperature (T_t) for checking aircraft air data systems and equipments.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Location and conditions for preproduction testing (see 4.5)

c. Sampling test requirements (see 4.6.2)

d. Level of preservation, packaging, and packing (see 5.1 and 5.2).

6.3 Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 4920-F290

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD, FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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